

SLAVE LAKE, AB, Canada

WMO: 713690

Lat: 55.2964N

Lon: 114.7844W

Elev: 583

StdP: 94.52

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-31.6	-28.5	-36.4	0.1	-31.4	-33.2	0.2	-28.2	14.5	2.5	13.1	2.5	2.6	120	0.694

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.9	26.3	17.4	24.3	16.5	22.6	15.8	18.8	24.2	17.8	22.8	16.7	21.4	4.5	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.7	12.8	21.5	15.6	11.9	20.5	14.7	11.2	19.4	56.0	24.5	52.4	22.9	49.2	21.5	22.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.6	10.0	8.6	-35.6	29.7	4.1	1.9	-38.5	31.0	-40.9	32.2	-43.2	33.2	-46.1	34.6		
(5)			-35.6	20.5	3.8	1.1	-38.3	21.3	-40.5	22.0	-42.7	22.6	-45.4	23.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	2.2	-12.9	-10.4	-5.3	2.6	9.3	13.9	16.5	15.2	10.2	3.6	-5.2	-12.0		
	DBStd	11.95	8.88	7.15	7.60	4.86	4.25	2.86	2.57	2.97	3.80	4.21	6.40	7.85		
	HDD10.0	3432	711	570	474	224	64	3	0	2	43	202	456	682		
	HDD18.3	5913	969	803	733	472	281	134	68	104	244	457	706	940		
	CDD10.0	581	0	0	0	2	42	121	201	162	49	3	0	0		
	CDD18.3	21	0	0	0	0	1	3	11	7	0	0	0	0		
	CDH23.3	275	0	0	0	2	19	43	116	80	16	0	0	0		
	CDH26.7	44	0	0	0	0	3	5	19	15	2	0	0	0		
(14) Wind	WSAvg	3.6	3.8	3.5	3.7	3.5	3.4	3.3	3.4	3.1	3.5	3.9	3.8	3.9		
(15) Precipitation	PrecAvg	525	27	18	21	21	42	92	100	69	52	29	20	29		
	PrecMax	659	69	42	59	44	108	186	197	137	132	70	46	61		
	PrecMin	349	8	5	2	0	15	11	40	6	17	2	1	5		
	PrecStd	81	15	11	12	14	24	48	41	36	28	17	13	15		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.8	7.1	10.6	21.2	25.8	27.0	28.8	28.6	25.5	18.6	10.4	8.0		
		MCWB	3.7	2.5	4.7	10.3	13.8	16.8	19.2	18.9	16.0	11.7	5.4	3.5		
	2%	DB	5.1	4.4	7.6	15.5	22.3	24.3	26.4	25.5	22.0	14.3	6.8	4.8		
		MCWB	2.0	1.1	3.1	7.6	12.1	15.3	18.2	18.0	14.9	8.6	2.9	1.4		
	5%	DB	2.5	2.1	5.9	12.3	19.7	22.3	24.2	23.2	19.4	12.2	4.8	1.9		
		MCWB	0.2	-0.4	2.0	5.7	11.1	14.6	17.4	17.0	13.7	7.3	1.8	-0.7		
	10%	DB	-0.7	-0.1	4.3	9.8	17.4	20.5	22.4	21.4	17.1	10.3	2.8	-1.0		
		MCWB	-2.3	-2.0	1.0	4.4	10.1	14.0	16.6	16.1	12.4	6.2	0.5	-2.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.9	3.1	5.2	10.6	14.7	18.1	20.5	20.2	17.4	12.2	5.7	3.8		
		MCDB	7.4	6.4	9.5	20.2	22.7	24.1	26.2	26.1	23.7	18.0	9.5	7.8		
	2%	WB	2.1	1.4	3.5	7.8	13.3	16.7	19.1	18.8	15.6	9.4	3.7	1.6		
		MCDB	4.8	4.1	7.2	14.4	20.3	22.2	24.6	24.1	20.8	13.2	6.4	4.7		
	5%	WB	0.3	-0.3	2.3	6.1	11.9	15.6	18.1	17.6	14.1	7.8	2.0	-0.5		
		MCDB	2.4	1.9	5.6	11.4	18.3	20.5	23.1	22.4	18.6	11.3	4.2	1.7		
	10%	WB	-2.3	-2.1	1.1	4.7	10.7	14.5	17.1	16.6	12.9	6.5	0.7	-2.5		
		MCDB	-0.7	-0.1	3.9	9.2	16.0	19.1	21.7	20.9	16.5	9.9	2.6	-0.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.9	10.3	10.3	10.3	11.6	10.3	9.9	10.3	10.4	8.6	7.4	8.4		
		MCDBR	11.8	13.0	11.6	15.7	16.1	14.7	13.9	14.0	15.3	12.9	9.7	12.4		
		MCWBR	9.2	10.2	7.8	9.2	8.2	7.6	7.2	7.7	9.0	8.2	7.0	9.6		
	5% WB	MCDBR	11.5	12.5	11.0	14.2	14.2	12.1	12.5	12.7	14.0	11.7	8.9	12.1		
		MCWBR	9.1	10.0	7.5	8.7	7.5	6.8	7.0	7.3	8.6	7.8	6.7	9.6		
(40) Clear-Sky Solar Irradiance	taub		0.247	0.261	0.289	0.333	0.353	0.362	0.367	0.364	0.319	0.288	0.254	0.230		
	taud		2.377	2.442	2.456	2.406	2.401	2.387	2.423	2.385	2.532	2.584	2.468	2.332		
	Ebn at Noon		722	849	897	893	891	882	871	849	853	809	714	656		
	Edh at Noon		48	66	83	101	108	111	105	103	78	58	44	39		
(44) All-Sky Solar Radiation	RadAvg		0.75	1.72	3.18	4.57	5.57	5.78	5.76	4.79	3.29	1.79	0.82	0.53		
	RadStd		0.07	0.16	0.31	0.38	0.48	0.43	0.34	0.36	0.31	0.23	0.08	0.04		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	+2.45	N/A	N/A	N/A	N/A	N/A	+55	N/A
(47)	Regional (23 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+63	N/A

Nomenclature: See separate page